Confidence-Aware Sparse-View 3D Reconstruction on Unseen Scenes

Project Proposal

Computer Vision Project

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Research objective

Improve sparse-view 3D reconstruction on unseen indoor scenes by reducing three common failure modes: far-reference views, severe occlusion, and repeated structures.

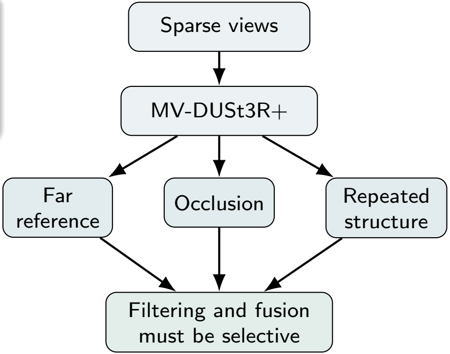
- Sparse-view settings provide limited geometric evidence and weaker co-visibility across images.
- Naive fusion of all predicted 3D points often introduces noise, structural inconsistency, and unstable point clouds.
- The project targets a modular extension to a pretrained reconstruction backbone rather than a new backbone architecture.

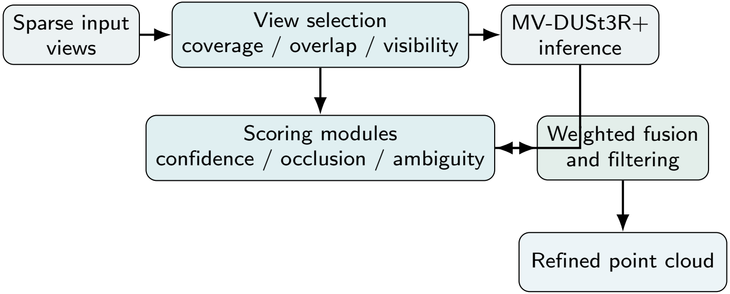
Baseline backbone

MV-DUSt3R+ is a single-stage multi-view reconstruction network designed for sparse-view scene reconstruction.

Observed limitations in sparse-view scenes

- **Far-reference views**: large viewpoint gaps reduce reliable correspondence.
- **Occlusion**: unsupported regions may still contribute to the final point cloud.
- **Repeated structures**: visually similar regions increase matching ambiguity.





Fusion weight

$$W_{\text{final}}(p) = C(p)\,(1 - O(p))\,S_{mv}(p)\,(1 - A(p))\,G(p)\,M(p)$$

- C: confidence, O: occlusion, S_{mv}: multi-view support
- A: ambiguity, G: anchor support, M: multi-view agreement

1. **Visibility-aware view selection**
   - Compare random, coverage-aware, diversity-aware, and hybrid view policies.
   - Reduce far-reference cases and improve co-visibility before inference.
2. **Occlusion-aware weighted fusion**
   - Suppress pixels that are weakly supported, poorly reprojected, or likely occluded.
   - Reduce unstable geometry in the final point cloud.
3. **Ambiguity-aware repeated-structure filtering**
   - Use ambiguity estimation, anchor pixels, and multi-view agreement for repeated indoor patterns.

Ambiguity estimation

$$A(p) = \alpha S(p) + \beta U(p) + \gamma R(p)$$

- $S(p)$: self-similarity within the image
- $U(p) = 1 - C(p)$: uncertainty from model confidence
- $R(p)$: reprojection disagreement across views

Decision principle A candidate region is preserved only when it satisfies:

- low ambiguity,
- support from reliable anchors,
- agreement across multiple views.

Targeted failure case

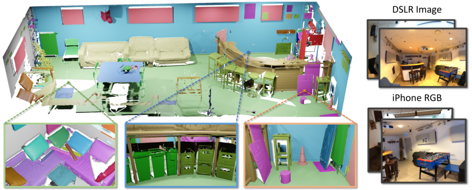
Repeated structures such as chairs, windows, and cabinet patterns may produce geometrically incorrect but visually plausible matches.

Expected effect

- fewer false correspondences,
- cleaner repeated-pattern regions,
- improved structural consistency.

Primary dataset: ScanNet++

- High-fidelity indoor-scene benchmark with registered DSLR images, iPhone RGB-D streams, and laser-scanned geometry.
- Appropriate for this project because it contains clutter, occlusion, and repeated indoor structures.
- Well suited for sparse-view experiments on unseen scenes.



Optional supplementary dataset

- A smaller subset from ScanNet or CO3Dv2 may be used for rapid prototyping and ablation.

Protocol

- Evaluate on unseen indoor scenes under 2-, 3-, and 5-view settings.
- Compare the baseline MV-DUSt3R+ with each proposed module and with the full pipeline.

Evaluation criteria

- reconstruction accuracy, completeness, and consistency,
- runtime per scene,
- case-based analysis on far-reference, occlusion-heavy, and repeated-structure scenarios.

Why the project is feasible

- The project builds on a pretrained backbone and introduces modular additions rather than a new backbone.
- Each module can be implemented and evaluated incrementally.

Nguyễn Ngọc Minh

- Literature review and related work
- Dataset preparation and experimental protocol

Kiều Sơn Tùng

- Backbone setup and baseline inference
- View-selection and scoring implementation

Đặng Thịnh Tường Minh

- Weighted fusion and repeated-structure module
- Evaluation, visualization, and report integration

Collaboration plan

Implementation, experiment design, and result discussion will be reviewed jointly across the group at each project milestone.

- A sparse-view reconstruction pipeline with improved robustness on unseen indoor scenes.
- Empirical evidence of how view selection, occlusion-aware fusion, and repeated-structure filtering affect reconstruction quality.
- A final implementation and report demonstrating both experimental rigor and computer vision methodology.

- Wang et al. *DUSt3R: Geometric 3D Vision Made Easy*. CVPR 2024.
- Tang et al. *MV-DUSt3R+: Single-Stage Scene Reconstruction from Sparse Views in 2 Seconds*. CVPR 2025.
- Yeshwanth et al. *ScanNet++: A High-Fidelity Dataset of 3D Indoor Scenes*. ICCV 2023.